

CounterPulse AI

Technology Stack Q&A; and End-to-End Workflow

Purpose	An evidence-first cyber-fraud incident-response platform that converts fragmented evidence into a reviewable case and human-approved recovery actions.
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Tech Stack Q&A;

1. What is the frontend stack?

React 19, TypeScript, Vite, Tailwind CSS 4, and Lucide React. React supplies reusable case-management views; TypeScript validates client-side data shapes; Vite enables fast development and builds; Tailwind creates consistent responsive styling; and Lucide provides clear UI icons.

2. Why does the backend use FastAPI and Uvicorn?

FastAPI provides typed REST APIs, async execution, WebSockets, and automatic API documentation. Uvicorn runs the ASGI service. Together they suit concurrent file processing, AI calls, and live voice sessions.

3. How are requests and settings validated?

Pydantic 2 validates API payloads and structured AI results. pydantic-settings reads environment-based configuration such as model keys, database URLs, storage paths, and email settings without embedding secrets in source code.

4. Which database is used?

SQLite is the default database, accessed through SQLAlchemy 2. It keeps the demo portable with no separate database server while SQLAlchemy's models preserve a clean route to a production database later.

5. How are evidence files handled safely?

Files are stored in local storage while metadata remains in the database. The backend validates MIME types and extensions, enforces a 25 MB limit, sanitizes filenames, blocks dangerous executables and path traversal, and calculates SHA-256 hashes.

6. What evidence formats are supported?

Images, PDFs, text/SMS, chat exports, audio, and URLs. A processor factory selects a dedicated processor for each type, keeping the evidence pipeline modular and extensible.

7. How are screenshots and scanned documents read?

RapidOCR with ONNX Runtime extracts text locally. Pillow decodes images and NumPy supplies pixel arrays to OCR. The app retains confidence, bounding-box references, and source evidence links.

8. How are PDFs processed?

pypdf extracts selectable text from multi-page PDFs. If a page is image-only or scanned, the application uses the OCR path as a fallback.

9. Which AI providers are supported?

Google Gemini is the default through the official google-genai SDK; the configured default model is gemini-2.0-flash. An OpenAI-compatible provider defaults to gpt-4o, and a deterministic mock provider supports offline tests.

10. Why use specialized AI agents?

The orchestrator coordinates evidence intelligence, incident reconstruction, triage, scam intelligence, compromise assessment, network intelligence, response generation, and follow-up. Focused agents give each task structured, traceable output.

11. How are indicators extracted?

Regex extraction finds structured values such as phone numbers, UPI IDs, URLs, emails, amounts, and transaction IDs. AI adds tactics, impersonated entities, urgency, and requested actions while preserving evidence provenance.

12. What is the Case Passport?

It is a consolidated, provenance-backed dossier containing the timeline, financial loss, indicators, evidence, risk assessment, and recovery steps. It turns scattered uploads into a coherent, auditable fraud case.

13. How are formal reports created and delivered?

ReportLab produces PDF dossiers and report packages such as bank-dispute and cybercrime complaint drafts. Resend, SMTP, and mock email providers are supported. A human approval gate is required before consequential email is sent.

14. How does voice control work?

The browser uses the Web Speech API for recognition and speech synthesis. FastAPI provides REST fallback plus a WebSocket endpoint for live voice state and responses. Voice commands cannot bypass approval for email dispatch.

15. Why use REST and WebSockets together?

REST suits ordinary case, evidence, analysis, and report actions. WebSockets provide immediate updates for a conversational voice session, avoiding polling while preserving REST as a reliable fallback.

16. How is quality verified?

pytest and pytest-asyncio cover processors, security, evidence APIs, reports, AI providers and agents, scam intelligence, Case Passport, voice control, and call flows.

17. How is a failed AI request handled?

The provider layer distinguishes configuration, quota, service-unavailability, and schema-validation failures. This allows the API to return an actionable error instead of inventing a successful analysis when a provider cannot complete the request.

18. Why are AI responses constrained by schemas?

Each AI agent is prompted to return JSON that conforms to a Pydantic schema, then the response is parsed and validated. Structured output makes downstream use safer for reports, case fields, and user-interface rendering.

19. What makes the evidence trail trustworthy?

Every evidence item can retain its original filename, MIME type, size, SHA-256 hash, normalized content, processing state, and provenance records. Extracted indicators can point back to their evidence item or message reference.

20. How is live text ingestion kept under user control?

The live-text endpoint requires an explicit user_authorized flag. Requests without that authorization are rejected, preventing the companion flow from silently ingesting conversations.

21. Why is CORS configured?

CORS middleware permits the separate React frontend to call the FastAPI backend during development and deployment. In a production deployment, its permissive allowed-origins setting should be narrowed to trusted frontend domains.

22. How can the application scale beyond the demo?

The modular processor factory, provider factory, SQLAlchemy ORM, and service boundaries allow individual layers to be replaced. Typical production upgrades are PostgreSQL, object storage, background queues, restricted CORS, authentication, and deployed secret management.

23. What role does httpx play?

httpx is used for asynchronous HTTP communication with OpenAI-compatible APIs. It supports timeouts and lets the backend classify network and server errors while preserving FastAPI's async workflow.

24. Why is a mock AI provider included?

The deterministic mock provider lets the test suite exercise agent orchestration and indicator extraction without external credentials, cost, rate limits, or nondeterministic model responses. It is intentionally for offline tests, not production analysis.

End-to-End Technology Workflow

1. React + TypeScript UI -> 2. FastAPI REST/WebSocket APIs -> 3. Pydantic validation and security checks -> 4. Modular evidence processors -> 5. SQLAlchemy + SQLite case persistence -> 6. Gemini/OpenAI/mock provider selection -> 7. Specialized-agent orchestration -> 8. Case Passport and ReportLab PDF package -> 9. Explicit human approval -> 10. Resend/SMTP dispatch with audit data.

Presentation summary

CounterPulse AI combines a React/TypeScript user experience, FastAPI services, SQL-backed evidence records, local OCR/PDF processing, configurable Gemini/OpenAI AI agents, and ReportLab/email integrations. Security validation, source provenance, and explicit human approval keep fraud-recovery actions traceable and under user control.